Math 54 — Week 7

Coordinate systems; change of basis; eigenvectors and eigenvalues; the characteristic equation (Lay 4.4, 4.6, 5.1–5.2)

§1 Facts used from Weeks 1–6

(W1) (Week 6) A basis of V is an independent spanning set; every basis of V has $\dim V$ vectors; in a space of dimension n, n independent vectors are a basis, and more than n vectors are dependent.

(W2) (Weeks 4–5) Invertible matrix theorem: for square A, invertible $\iff$ $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\iff$ $\det A \neq 0$ $\iff$ the columns form a basis of $\mathbb{R}^n$. For 2×2 matrices, $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

(W3) (Weeks 4–5) $\det(AB) = \det A \det B$, $\det A^T = \det A$, and the determinant of a triangular matrix is the product of its diagonal entries.

(W4) (Weeks 5–6) $\operatorname{Nul} A$ is a subspace, with a basis from the parametric vector form of $A\mathbf{x} = \mathbf{0}$. Row operations do not change the linear relations among the columns of a matrix (the proof of Lay 4.3, Thm. 6).

(W5) (Weeks 3, 5) A linear T is one-to-one if and only if $T(\mathbf{x}) = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$; the proof uses only linearity, so it holds for linear maps between any vector spaces.

(W6) (Week 6) If $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is dependent and $\mathbf{v}_1 \neq \mathbf{0}$, some $\mathbf{v}_j$ $(j > 1)$ is a combination of $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$ (Lay 4.3, Thm. 4).

§2 This week

§2.1 Coordinate systems (Lay 4.4)

The question. With the basis $\mathbf{b}_1 = (2, 1)$, $\mathbf{b}_2 = (-1, 1)$ of $\mathbb{R}^2$, how do we describe $\mathbf{x} = (4, 5)$ by a list of numbers relative to this basis, why is the list unique, and how do we compute it?

Theorem (The unique representation theorem (Lay 4.4, Thm. 7))
Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis for a vector space V. Then for each $\mathbf{x}$ in V there is a unique set of scalars $c_1, \ldots, c_n$ with $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$.

Proof. 1. $\mathcal{B}$ spans V, so there is at least one such list.

2. If also $\mathbf{x} = d_1\mathbf{b}_1 + \cdots + d_n\mathbf{b}_n$, subtracting gives $\mathbf{0} = (c_1 - d_1)\mathbf{b}_1 + \cdots + (c_n - d_n)\mathbf{b}_n$. $\mathcal{B}$ is independent, so $c_i - d_i = 0$ for every i.

□

Definition. These weights are the **coordinates of $\mathbf{x}$ relative to $\mathcal{B}$**, and $[\mathbf{x}]_{\mathcal{B}} = (c_1, \ldots, c_n)$ in $\mathbb{R}^n$ is the **$\mathcal{B}$-coordinate vector** of $\mathbf{x}$. The map $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is the **coordinate mapping** determined by $\mathcal{B}$.

Coordinates in $\mathbb{R}^n$. Let $P_{\mathcal{B}} = [\,\mathbf{b}_1 \cdots \mathbf{b}_n\,]$, the **change-of-coordinates matrix** from $\mathcal{B}$ to the standard basis. By the definition of a matrix times a vector,

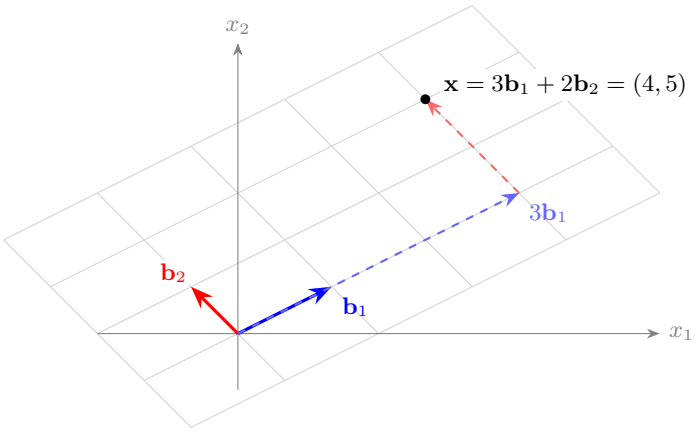
$$\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}, \qquad \text{so} \qquad [\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}.$$

$P_{\mathcal{B}}$ is invertible because its columns are a basis of $\mathbb{R}^n$ (W2). In practice, find $[\mathbf{x}]_{\mathcal{B}}$ by row reducing $[\,\mathbf{b}_1 \cdots \mathbf{b}_n \;\; \mathbf{x}\,]$.

The question answered. Solve $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 = \mathbf{x}$; interchange first to avoid fractions:

$$\left[\begin{array}{cc|c} 2 & -1 & 4 \\ 1 & 1 & 5 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & -1 & 4 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -6 \end{array}\right] \xrightarrow[R_1 - R_2]{-\frac{1}{3}R_2} \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 2 \end{array}\right], \qquad [\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Check: $3(2,1) + 2(-1,1) = (4,5)$. The list is unique by Theorem 7.



The grid of the basis $\mathcal{B}$: from $\mathbf{0}$ go 3 steps along $\mathbf{b}_1$, then 2 steps along $\mathbf{b}_2$. In standard coordinates the same point is $(4,5)$.

Theorem (The coordinate mapping (Lay 4.4, Thm. 8))
Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis for V. Then $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

Proof. 1. **Linear.** If $\mathbf{u} = \sum c_i\mathbf{b}_i$ and $\mathbf{w} = \sum d_i\mathbf{b}_i$, then $\mathbf{u} + \mathbf{w} = \sum(c_i + d_i)\mathbf{b}_i$ and $r\mathbf{u} = \sum(rc_i)\mathbf{b}_i$. By uniqueness these weights are the coordinates, so $[\,\mathbf{u} + \mathbf{w}\,]_{\mathcal{B}} = [\,\mathbf{u}\,]_{\mathcal{B}} + [\,\mathbf{w}\,]_{\mathcal{B}}$ and $[\,r\mathbf{u}\,]_{\mathcal{B}} = r[\,\mathbf{u}\,]_{\mathcal{B}}$.

2. **One-to-one.** If $[\,\mathbf{x}\,]_{\mathcal{B}} = \mathbf{0}$, then $\mathbf{x} = 0\mathbf{b}_1 + \cdots + 0\mathbf{b}_n = \mathbf{0}$. By (W5) the map is one-to-one.

3. **Onto.** Each $(c_1, \ldots, c_n)$ in $\mathbb{R}^n$ is the coordinate vector of $c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$.

□

A one-to-one linear map from V onto W is an **isomorphism**. By Theorem 8, every vector space with a basis of n vectors is isomorphic to $\mathbb{R}^n$.

Coordinates keep linear relations. For $\mathbf{u}_1, \ldots, \mathbf{u}_p$ in V and scalars c_j, linearity of the coordinate map (Theorem 8) gives

$$c_1[\,\mathbf{u}_1\,]_{\mathcal{B}} + \cdots + c_p[\,\mathbf{u}_p\,]_{\mathcal{B}} = [\,c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p\,]_{\mathcal{B}},$$

and a coordinate vector is $\mathbf{0}$ only for the vector $\mathbf{0}$. So

$$c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p = \mathbf{0} \quad \Longleftrightarrow \quad c_1[\,\mathbf{u}_1\,]_{\mathcal{B}} + \cdots + c_p[\,\mathbf{u}_p\,]_{\mathcal{B}} = \mathbf{0}.$$

Hence $\{\mathbf{u}_j\}$ is independent exactly when $\{[\,\mathbf{u}_j\,]_{\mathcal{B}}\}$ is, and in the same way $\mathbf{y}$ is in $\mathrm{Span}\{\mathbf{u}_j\}$ exactly when $[\,\mathbf{y}\,]_{\mathcal{B}}$ is in $\mathrm{Span}\{[\,\mathbf{u}_j\,]_{\mathcal{B}}\}$. Span and independence questions in V become the same questions about columns of numbers.

Worked example: independence of polynomials. Are $p_1 = 1+2t^2$, $p_2 = 4+t+5t^2$, $p_3 = 3+2t$ independent in $\mathbb{P}_2$? With the standard basis $\{1, t, t^2\}$ their coordinate vectors are $(1,0,2)$, $(4,1,5)$, $(3,2,0)$. Row reduce the matrix with these columns:

$$\begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 2 \\ 2 & 5 & 0 \end{bmatrix} \xrightarrow{R_3-2R_1} \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 2 \\ 0 & -3 & -6 \end{bmatrix} \xrightarrow{R_3+3R_2} \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1-4R_2} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Column 3 of the reduced form is -5 times column 1 plus 2 times column 2. Row operations keep column relations (W4), and coordinates keep relations, so

$$p_3 = -5p_1 + 2p_2 : \qquad -5(1+2t^2) + 2(4+t+5t^2) = 3+2t.$$

The polynomials are dependent.

Worked example: coordinates on a subspace. Let $\mathbf{v}_1 = (1,2,0)$, $\mathbf{v}_2 = (0,1,1)$, $H = \mathrm{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ (a plane in $\mathbb{R}^3$), and $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$, a basis of H (two vectors, not multiples). Is $\mathbf{x} = (2,7,3)$ in H, and if so, what is $[\mathbf{x}]_{\mathcal{B}}$?

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 2 & 1 & 7 \\ 0 & 1 & 3 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{array}\right] \xrightarrow{R_3-R_2} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{array}\right]$$

The system is consistent, so $\mathbf{x} \in H$, and $[\mathbf{x}]_{\mathcal{B}} = (2,3)$: a vector in $\mathbb{R}^2$, although $\mathbf{x}$ is in $\mathbb{R}^3$. Check: $2(1,2,0) + 3(0,1,1) = (2,7,3)$.

Shortcuts

- **Coordinates in $\mathbb{R}^n$**: row reduce $[\,\mathbf{b}_1 \cdots \mathbf{b}_n \;\; \mathbf{x}\,]$. Reason: $\mathbf{x} = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$. Instance: the question.
- **Questions in $\mathbb{P}_n$ or matrix spaces**: pass to coordinates in the standard basis and work with columns of numbers. Reason: coordinates keep linear relations. Instance: the polynomial example.
- **Coordinates of a polynomial in a non-standard basis**: match the coefficients of $1, t, t^2$ to get a linear system for the weights. Reason: two polynomials are equal exactly when their coefficients are equal. Instance: in $\mathbb{P}_1$ with $\mathcal{B} = \{1+t,\ 1-t\}$, $c_1(1+t) + c_2(1-t) = 5+t$ gives

  $$c_1 + c_2 = 5, \qquad c_1 - c_2 = 1,$$

  so $c_1 = 3$, $c_2 = 2$ and $[5+t]_{\mathcal{B}} = (3,2)$.

Traps

- The order of the basis matters: with $\mathcal{B}' = \{\mathbf{b}_2, \mathbf{b}_1\}$, the same $\mathbf{x} = (4,5)$ has $[\mathbf{x}]_{\mathcal{B}'} = (2,3)$, not $(3,2)$.
- List polynomial coefficients in the order of the basis $\{1, t, t^2\}$: $1+2t^2$ has coordinates $(1,0,2)$, not $(2,0,1)$.
- $P_{\mathcal{B}}$ turns $\mathcal{B}$-coordinates into standard coordinates, not the reverse: in the question $P_{\mathcal{B}}(4,5) = (3,9)$, which is not $[\mathbf{x}]_{\mathcal{B}} = (3,2)$. The correct formula is $[\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}$.
- For a basis of a subspace H, $[\mathbf{x}]_{\mathcal{B}}$ has $\dim H$ entries, not n: in the subspace example it is in $\mathbb{R}^2$.

§2.2 Change of basis (Lay 4.6)

The question. Let $\mathcal{B} = \{(1,1),(1,2)\}$ and $\mathcal{C} = \{(1,0),(2,1)\}$. A vector has $\mathcal{B}$-coordinates $(2,-1)$. How do we find its $\mathcal{C}$-coordinates directly, without converting to standard coordinates first?

Theorem (Change of coordinates (Lay 4.6, Thm. 15))

Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \ldots, \mathbf{c}_n\}$ be bases of V. Then there is a unique $n \times n$ matrix $P_{\mathcal{C}\leftarrow\mathcal{B}}$ with $[\mathbf{x}]_\mathcal{C} = P_{\mathcal{C}\leftarrow\mathcal{B}}[\mathbf{x}]_\mathcal{B}$ for every $\mathbf{x}$ in V. Its columns are the $\mathcal{C}$-coordinate vectors of the vectors of $\mathcal{B}$:

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix}[\mathbf{b}_1]_\mathcal{C} & [\mathbf{b}_2]_\mathcal{C} & \cdots & [\mathbf{b}_n]_\mathcal{C}\end{bmatrix}.$$

It is invertible, and $(P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1} = P_{\mathcal{B}\leftarrow\mathcal{C}}$.

The idea

Taking $\mathcal{C}$-coordinates is linear, so it is enough to know the $\mathcal{C}$-coordinates of the vectors $\mathbf{b}_j$: those of any $\mathbf{x} = \sum c_j\mathbf{b}_j$ are the same combination. The *old* basis vectors written in the *new* basis are the columns.

Proof. 1. Write $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$, so $[\mathbf{x}]_\mathcal{B} = (c_1, \ldots, c_n)$. The $\mathcal{C}$-coordinate map is linear (Theorem 8), so

$$[\mathbf{x}]_\mathcal{C} = c_1[\mathbf{b}_1]_\mathcal{C} + \cdots + c_n[\mathbf{b}_n]_\mathcal{C} = P_{\mathcal{C}\leftarrow\mathcal{B}}[\mathbf{x}]_\mathcal{B}.$$

2. Uniqueness: if a matrix M also works, apply both to $[\mathbf{b}_j]_\mathcal{B} = \mathbf{e}_j$: column j of M is $M\mathbf{e}_j = [\mathbf{b}_j]_\mathcal{C}$.
3. Invertible: if $\sum_j d_j[\mathbf{b}_j]_\mathcal{C} = \mathbf{0}$, then $[\sum d_j\mathbf{b}_j]_\mathcal{C} = \mathbf{0}$, so $\sum d_j\mathbf{b}_j = \mathbf{0}$ and every $d_j = 0$ since $\mathcal{B}$ is independent. So the columns are independent and $P_{\mathcal{C}\leftarrow\mathcal{B}}$ is invertible (W2).
4. Multiplying step 1 by the inverse gives $[\mathbf{x}]_\mathcal{B} = (P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}[\mathbf{x}]_\mathcal{C}$ for all $\mathbf{x}$. Apply this to $\mathbf{x} = \mathbf{c}_j$, whose $\mathcal{C}$-coordinate vector is $\mathbf{e}_j$: column j of $(P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}$ is $(P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}\mathbf{e}_j = [\mathbf{c}_j]_\mathcal{B}$, which is column j of $P_{\mathcal{B}\leftarrow\mathcal{C}}$. So $(P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1} = P_{\mathcal{B}\leftarrow\mathcal{C}}$.

□

Computing $P_{\mathcal{C}\leftarrow\mathcal{B}}$ in $\mathbb{R}^n$. Column j, $[\mathbf{b}_j]_\mathcal{C}$, solves $[\mathbf{c}_1 \cdots \mathbf{c}_n]\mathbf{y} = \mathbf{b}_j$. Solve all of them at once:

$$[\,\mathbf{c}_1 \;\cdots\; \mathbf{c}_n \;\mid\; \mathbf{b}_1 \;\cdots\; \mathbf{b}_n\,] \longrightarrow [\,I \;\mid\; P_{\mathcal{C}\leftarrow\mathcal{B}}\,].$$

The question answered. With $\mathcal{B} = \{(1,1),(1,2)\}$ and $\mathcal{C} = \{(1,0),(2,1)\}$:

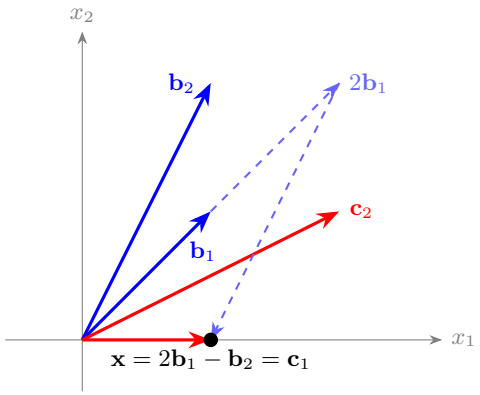
$$\left[\begin{array}{cc|cc}1 & 2 & 1 & 1\\ 0 & 1 & 1 & 2\end{array}\right] \xrightarrow{R_1-2R_2} \left[\begin{array}{cc|cc}1 & 0 & -1 & -3\\ 0 & 1 & 1 & 2\end{array}\right], \qquad P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix}-1 & -3\\ 1 & 2\end{bmatrix}.$$

Check the first column: $-1(1,0) + 1(2,1) = (1,1) = \mathbf{b}_1$. Then

$$[\mathbf{x}]_\mathcal{C} = \begin{bmatrix}-1 & -3\\ 1 & 2\end{bmatrix}\begin{bmatrix}2\\ -1\end{bmatrix} = \begin{bmatrix}-2+3\\ 2-2\end{bmatrix} = \begin{bmatrix}1\\ 0\end{bmatrix}.$$

Check: $\mathbf{x} = 2(1,1) - (1,2) = (1,0)$, which is $1\cdot(1,0) + 0\cdot(2,1)$. The other direction uses the 2×2 inverse formula (determinant $-2+3=1$):

$$P_{\mathcal{B}\leftarrow\mathcal{C}} = (P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1} = \begin{bmatrix}2 & 3\\ -1 & -1\end{bmatrix}.$$



The question's vector: $[\mathbf{x}]_\mathcal{B} = (2,-1)$, and $\mathbf{x} = (1,0)$ is $\mathbf{c}_1$ itself, so $[\mathbf{x}]_\mathcal{C} = (1,0)$.

- With $\mathcal{C}$ the standard basis $\mathcal{E}$, $P_{\mathcal{E}\leftarrow\mathcal{B}} = P_\mathcal{B}$, because $[\mathbf{b}_j]_\mathcal{E} = \mathbf{b}_j$.
- In $\mathbb{R}^n$, $P_{\mathcal{C}\leftarrow\mathcal{B}} = P_\mathcal{C}^{-1}P_\mathcal{B}$: go from $\mathcal{B}$ to standard coordinates, then from standard to $\mathcal{C}$.

Worked example: abstract bases. In some space V, $\mathbf{b}_1 = 2\mathbf{c}_1 - \mathbf{c}_2$ and $\mathbf{b}_2 = \mathbf{c}_1 + 3\mathbf{c}_2$. For $\mathbf{x} = 3\mathbf{b}_1 + \mathbf{b}_2$,

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}, \qquad [\,\mathbf{x}\,]_{\mathcal{C}} = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix}.$$

Check: $3(2\mathbf{c}_1 - \mathbf{c}_2) + (\mathbf{c}_1 + 3\mathbf{c}_2) = 7\mathbf{c}_1$.

Worked example: change of basis in $\mathbb{P}_2$. Let $\mathcal{B} = \{1,\ 1+t,\ 1+t+t^2\}$ and $\mathcal{C} = \{1, t, t^2\}$. The columns of $P_{\mathcal{C}\leftarrow\mathcal{B}}$ are the coefficient vectors of the $\mathcal{B}$ polynomials, and its inverse can be checked by multiplying:

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad P_{\mathcal{B}\leftarrow\mathcal{C}} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

For $p = 2 + 3t + 4t^2$,

$$[\,p\,]_{\mathcal{B}} = P_{\mathcal{B}\leftarrow\mathcal{C}} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2-3 \\ 3-4 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 4 \end{bmatrix}.$$

Check: $-1 - (1+t) + 4(1+t+t^2) = 2 + 3t + 4t^2$.

Shortcuts

- **Change of basis in $\mathbb{R}^n$**: row reduce $[\,\mathcal{C} \mid \mathcal{B}\,]$ to $[\,I \mid P_{\mathcal{C}\leftarrow\mathcal{B}}\,]$. Reason: it solves the n systems $[\,\mathbf{c}_1 \cdots \mathbf{c}_n\,]\mathbf{y} = \mathbf{b}_j$ at once. Instance: the question.
- **The other direction**: $P_{\mathcal{B}\leftarrow\mathcal{C}} = (P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}$; for 2×2 use the inverse formula. Reason: Theorem 15. Instance: the question.
- **Bases given in terms of each other**: the coefficients of $\mathbf{b}_j$ in terms of $\mathcal{C}$ form column j. Reason: Theorem 15. Instance: the abstract example.

Traps

- The order matters: $[\,\mathcal{B} \mid \mathcal{C}\,]$ produces $P_{\mathcal{B}\leftarrow\mathcal{C}}$, the inverse of the matrix wanted. For the question,

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & -1 \end{array}\right],$$

  which is $P_{\mathcal{B}\leftarrow\mathcal{C}}$, not $P_{\mathcal{C}\leftarrow\mathcal{B}}$.
- In the abstract example, the coefficients of $\mathbf{b}_1$ go into a *column*: writing them as row 1 gives the transpose, which sends $(3,1)$ to $(5,6)$, not $(7,0)$.

§2.3 Eigenvectors and eigenvalues (Lay 5.1)

The question. The matrix below sends $(3,-2)$ to $(-9,11)$, off the line through $(3,-2)$, but it sends $(6,-5)$ to $(-24,20) = -4(6,-5)$, on its own line. Which vectors does a matrix only stretch, and how do they help to compute $A^k\mathbf{x}$?

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$$

Definition. An **eigenvector** of an $n \times n$ matrix A is a *nonzero* vector $\mathbf{x}$ with $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. A scalar λ is an **eigenvalue** of A if $A\mathbf{x} = \lambda\mathbf{x}$ has a nontrivial solution; such an $\mathbf{x}$ is an eigenvector **corresponding to** λ. The set of all solutions of $(A - \lambda I)\mathbf{x} = \mathbf{0}$ is the **eigenspace** of A for λ.

- λ **is an eigenvalue** $\iff$ $(A-\lambda I)\mathbf{x}=\mathbf{0}$ **has a nontrivial solution**, because $A\mathbf{x}=\lambda\mathbf{x}$ is the same equation as $A\mathbf{x}-\lambda I\mathbf{x}=\mathbf{0}$.
- The eigenspace is $\mathrm{Nul}(A-\lambda I)$, a subspace: $\mathbf{0}$ together with all eigenvectors for λ.
- **Checking a vector**: compute $A\mathbf{u}$ and see whether it is a multiple of $\mathbf{u}$. **Checking a number**: row reduce $A-\lambda I$ and look for a free variable.

Worked example: an eigenspace. Is $\lambda=2$ an eigenvalue of the matrix below, and what is its eigenspace?

$$A=\begin{bmatrix}4&-1&6\\2&1&6\\2&-1&8\end{bmatrix},\qquad A-2I=\begin{bmatrix}2&-1&6\\2&-1&6\\2&-1&6\end{bmatrix}\xrightarrow[R_3-R_1]{R_2-R_1}\begin{bmatrix}2&-1&6\\0&0&0\\0&0&0\end{bmatrix}.$$

1. Two free variables, so $(A-2I)\mathbf{x}=\mathbf{0}$ has nontrivial solutions and 2 is an eigenvalue.
2. The equation is $x_1=\frac{1}{2}x_2-3x_3$, so
$$\mathbf{x}=x_2\begin{bmatrix}1/2\\1\\0\end{bmatrix}+x_3\begin{bmatrix}-3\\0\\1\end{bmatrix},$$
and a basis of the eigenspace is $\{(1,2,0),(-3,0,1)\}$ (the first vector doubled).
3. A multiplies every vector of this plane by 2.

Theorem (Triangular matrices (Lay 5.1, Thm. 1))
The eigenvalues of a triangular matrix are the entries on its main diagonal.

Proof. 1. If A is triangular, so is $A-\lambda I$, with diagonal entries $a_{11}-\lambda,\dots,a_{nn}-\lambda$.

2. λ is an eigenvalue exactly when $(A-\lambda I)\mathbf{x}=\mathbf{0}$ has a nontrivial solution, that is, when $A-\lambda I$ is not invertible, that is, when $\det(A-\lambda I)=0$ (W2).

3. By (W3), $\det(A-\lambda I)=(a_{11}-\lambda)\cdots(a_{nn}-\lambda)$, which is 0 exactly when $\lambda=a_{ii}$ for some i.

□

0 **is an eigenvalue of** A $\iff$ A **is not invertible**, because $A\mathbf{x}=0\mathbf{x}$ is $A\mathbf{x}=\mathbf{0}$.

Theorem (Distinct eigenvalues (Lay 5.1, Thm. 2))
If $\mathbf{v}_1,\dots,\mathbf{v}_r$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1,\dots,\lambda_r$ of an $n\times n$ matrix A, then $\{\mathbf{v}_1,\dots,\mathbf{v}_r\}$ is linearly independent.

The idea

For two vectors: if $\mathbf{v}_2=c\mathbf{v}_1$, applying A gives $\lambda_2\mathbf{v}_2=c\lambda_1\mathbf{v}_1$, while multiplying by λ_2 gives $\lambda_2\mathbf{v}_2=c\lambda_2\mathbf{v}_1$. Subtracting, $c(\lambda_1-\lambda_2)\mathbf{v}_1=\mathbf{0}$, so $c=0$ and $\mathbf{v}_2=\mathbf{0}$, which is impossible. The same subtraction, applied to the first vector that is a combination of earlier ones, gives the general case.

Proof. 1. Suppose the set is dependent. $\mathbf{v}_1\neq\mathbf{0}$, so by (W6) some $\mathbf{v}_j$ is a combination of the vectors before it; let $\mathbf{v}_{p+1}$ be the first such vector ($p\geq 1$). Then $\{\mathbf{v}_1,\dots,\mathbf{v}_p\}$ is independent, since otherwise by (W6) an earlier vector would already be a combination of the ones before it. Write

$$\mathbf{v}_{p+1}=c_1\mathbf{v}_1+\cdots+c_p\mathbf{v}_p.$$

2. Multiply by A, using $A\mathbf{v}_i=\lambda_i\mathbf{v}_i$:
$$\lambda_{p+1}\mathbf{v}_{p+1}=c_1\lambda_1\mathbf{v}_1+\cdots+c_p\lambda_p\mathbf{v}_p.$$

3. Multiply step 1 by λ_{p+1}:
$$\lambda_{p+1}\mathbf{v}_{p+1}=c_1\lambda_{p+1}\mathbf{v}_1+\cdots+c_p\lambda_{p+1}\mathbf{v}_p.$$

4. Subtract: $\mathbf{0} = c_1(\lambda_1 - \lambda_{p+1})\mathbf{v}_1 + \cdots + c_p(\lambda_p - \lambda_{p+1})\mathbf{v}_p$. By independence each $c_i(\lambda_i - \lambda_{p+1}) = 0$, and $\lambda_i \neq \lambda_{p+1}$, so each $c_i = 0$.
5. Then $\mathbf{v}_{p+1} = \mathbf{0}$, which an eigenvector is not. So the set is independent.

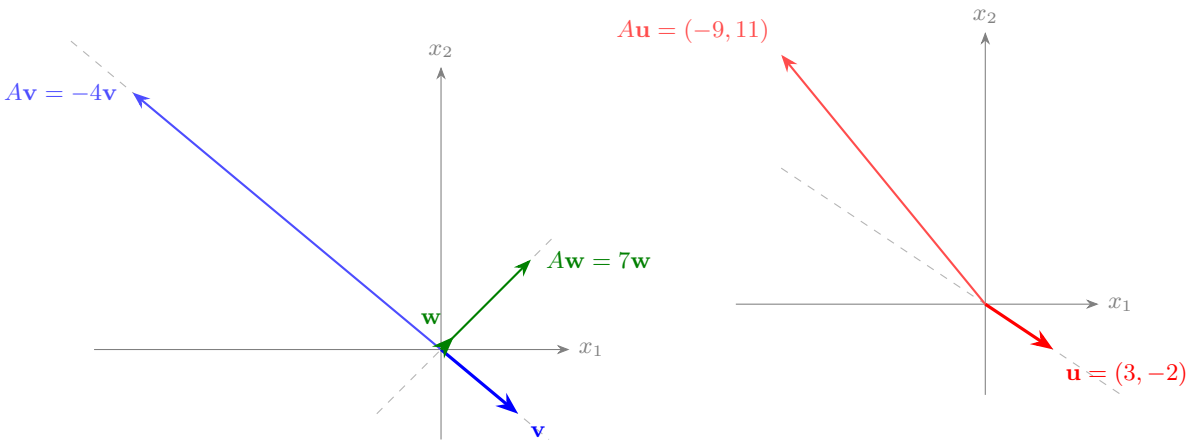
□

An $n \times n$ matrix has at most n distinct eigenvalues: eigenvectors for $n+1$ distinct eigenvalues would be $n+1$ independent vectors in $\mathbb{R}^n$, which is impossible (W1).

Eigenvectors and powers. If $A\mathbf{v} = \lambda\mathbf{v}$, then $A^2\mathbf{v} = A(\lambda\mathbf{v}) = \lambda^2\mathbf{v}$, and repeating, $A^k\mathbf{v} = \lambda^k\mathbf{v}$. If $\mathbf{x}_0 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ with eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ for λ_1, λ_2, then by linearity

$$A^k\mathbf{x}_0 = c_1\lambda_1^k\mathbf{v}_1 + c_2\lambda_2^k\mathbf{v}_2.$$

This describes the sequence $\mathbf{x}_{k+1} = A\mathbf{x}_k$ (a **difference equation**) for all k at once.



The question's matrix. Left: the eigenvectors $\mathbf{v} = (6, -5)$ and $\mathbf{w} = (1, 1)$ stay on their lines (dashed) and are multiplied by -4 and 7. Right, at a larger scale: $\mathbf{u} = (3, -2)$ is not an eigenvector, and $A\mathbf{u}$ leaves the dashed line through $\mathbf{u}$.

The question answered. The vectors that A only stretches are the eigenvectors. Here $A(1, 1) = (1+6,\ 5+2) = (7, 7)$ and $A(6, -5) = -4(6, -5)$, so 7 and -4 are eigenvalues. Since $(7, -4) = (1, 1) + (6, -5)$,

$$A^k \begin{bmatrix} 7 \\ -4 \end{bmatrix} = 7^k \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-4)^k \begin{bmatrix} 6 \\ -5 \end{bmatrix}$$

for every k, with no matrix powers computed.

Shortcuts

- **Is u an eigenvector?** One multiplication: compare $A\mathbf{u}$ with $\mathbf{u}$. **Is λ an eigenvalue?** One row reduction of $A - \lambda I$, looking for a free variable. Reason: the first item of 5.1. Instance: the question, and $\lambda = 2$ above.
- **Triangular**: the eigenvalues are the diagonal entries; no determinant needed. Reason: Theorem 1. Instance: the matrix below has eigenvalues 2 and -1.

$$\begin{bmatrix} 2 & 5 \\ 0 & -1 \end{bmatrix}$$

- **Equal row sums**: if every row of A adds up to s, then s is an eigenvalue with eigenvector $(1, \ldots, 1)$. Reason: entry i of $A(1, \ldots, 1)$ is the sum of row i. Instance: the question's rows add up to 7.
- **Singular matrices**: 0 is an eigenvalue exactly when A is not invertible. Instance: the matrix below sends $(2, -1)$ to $\mathbf{0}$, so 0 is an eigenvalue.

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

- **Powers and inverses**: if $A\mathbf{x} = \lambda\mathbf{x}$, then $A^k\mathbf{x} = \lambda^k\mathbf{x}$, and if A is invertible, $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$ (multiply $A\mathbf{x} = \lambda\mathbf{x}$ by $\lambda^{-1}A^{-1}$; here $\lambda \neq 0$). Instance: $A^2(6,-5) = 16(6,-5)$.

Traps

- An eigenvector must be nonzero ($A\mathbf{0} = \lambda\mathbf{0}$ for every λ); an eigenvalue may be 0.
- The eigenspace is $\mathrm{Nul}(A - \lambda I)$; row reduce $A - \lambda I$, not A.
- Row reducing A does not preserve its eigenvalues. The question's matrix has eigenvalues 7 and -4, but $R_2 - 5R_1$ turns it into a triangular matrix with eigenvalues 1 and -28:

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \xrightarrow{R_2 - 5R_1} \begin{bmatrix} 1 & 6 \\ 0 & -28 \end{bmatrix}.$$

- Two independent eigenvectors need not have different eigenvalues: $(1,2,0)$ and $(-3,0,1)$ in the eigenspace example both have eigenvalue 2. Theorem 2 goes only one way.

§2.4 The characteristic equation (Lay 5.2)

The question. Checking one candidate λ at a time is slow. How do we find all eigenvalues of the matrix below at once, together with their eigenvectors?

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Theorem (The characteristic equation (Lay 5.2))
A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies $\det(A - \lambda I) = 0$.

Proof. 1. $A\mathbf{x} = \lambda\mathbf{x}$ is the same equation as $(A - \lambda I)\mathbf{x} = \mathbf{0}$, so λ is an eigenvalue exactly when this homogeneous system has a nontrivial solution.

2. By the invertible matrix theorem (W2), $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution exactly when $A - \lambda I$ is not invertible, which happens exactly when $\det(A - \lambda I) = 0$.

□

For an $n \times n$ matrix, $\det(A - \lambda I)$ is a polynomial in λ of degree n with leading coefficient $(-1)^n$, the **characteristic polynomial** (Lay states this without proof; it is cited here). The **(algebraic) multiplicity** of an eigenvalue is its multiplicity as a root. The **trace** $\operatorname{tr} A$ is the sum of the diagonal entries.

The 2×2 case. Expanding,

$$\det \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix} = (a - \lambda)(d - \lambda) - bc = \lambda^2 - (a + d)\lambda + (ad - bc) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$

If this has real roots λ_1, λ_2, then it equals $(\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2$; comparing coefficients, $\lambda_1 + \lambda_2 = \operatorname{tr} A$ and $\lambda_1\lambda_2 = \det A$. If the discriminant $(\operatorname{tr} A)^2 - 4 \det A$ is negative, there are no real eigenvalues.

The question answered. $\operatorname{tr} A = 6$ and $\det A = 8$, so

$$\det(A - \lambda I) = \lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4).$$

The eigenvalues are 2 and 4. For each, $A - \lambda I$ is singular, so its second row is a multiple of its first, and the first row's equation gives the eigenvectors:

$$\lambda = 2: \quad A - 2I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad x_1 + x_2 = 0, \quad \text{eigenvector } (-1,1);$$

$$\lambda = 4: \quad A - 4I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \quad -x_1 + x_2 = 0, \quad \text{eigenvector } (1,1).$$

Check: the eigenvalues add up to $6 = \operatorname{tr} A$ and multiply to $8 = \det A$.

Worked example (3×3). Expand $\det(A - \lambda I)$ down column 1, which has one nonzero entry, and keep the factors:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 0 \\ 0 & 1 & 2 \end{bmatrix}, \qquad \det(A - \lambda I) = (1-\lambda) \det \begin{bmatrix} 3-\lambda & 0 \\ 1 & 2-\lambda \end{bmatrix} = (1-\lambda)(3-\lambda)(2-\lambda).$$

The eigenvalues are $1, 2, 3$, each of multiplicity one.

More consequences.

- $\det A$ **is the product of the eigenvalues** when the characteristic polynomial factors as $(\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$ (eigenvalues repeated by multiplicity): set $\lambda = 0$.
- A **and** A^T **have the same eigenvalues**: $\det(A^T - \lambda I) = \det\big((A - \lambda I)^T\big) = \det(A - \lambda I)$ by (W3).
- **The invertible matrix theorem, continued**: A is invertible $\iff$ 0 is not an eigenvalue of A.

Definition. A and B ($n \times n$) are **similar** if $B = P^{-1}AP$ for some invertible P. Changing A into $P^{-1}AP$ is a **similarity transformation**.

Theorem (Similar matrices (Lay 5.2, Thm. 4))
If A and B are similar, they have the same characteristic polynomial, hence the same eigenvalues with the same multiplicities.

Proof. 1. $B - \lambda I = P^{-1}AP - \lambda P^{-1}P = P^{-1}(A - \lambda I)P$.

2. By the product rule (W3),

$$\det(B - \lambda I) = \det(P^{-1}) \det(A - \lambda I) \det P = \det(A - \lambda I),$$

because $\det(P^{-1}) \det P = \det(P^{-1}P) = 1$.

□

Shortcuts

- 2×2: $\det(A - \lambda I) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A$; real eigenvalues sum to the trace and multiply to the determinant. Reason and instance: above.
- 2×2 **eigenvector**: if $b \neq 0$, an eigenvector for λ is $(b,\ \lambda - a)$. Reason: $A - \lambda I$ is singular and its first row $(a - \lambda,\ b)$ is nonzero, so the second row is a multiple of the first, and $(a - \lambda)b + b(\lambda - a) = 0$. Condition: $b \neq 0$. Instance: the question with $\lambda = 4$ gives $(1, 1)$.
- 3×3 **characteristic polynomial**: expand $\det(A - \lambda I)$ along a row or column with zeros, and *keep factors* such as $(1 - \lambda)$ instead of multiplying out. Reason: the roots can then be read off. Instance: the 3×3 example.
- **Check eigenvalues with trace and determinant**. Reason: the comparison of coefficients above. Instance: the question.

Traps

- The equation is $\det(A - \lambda I) = 0$, not $\det A - \lambda = 0$: for the question's matrix, $\det A - \lambda = 8 - \lambda$ would give the wrong eigenvalue 8.
- A real matrix may have no real eigenvalues. The matrix below has $\operatorname{tr} = 4$, $\det = 5$, and $\lambda^2 - 4\lambda + 5$ has discriminant $16 - 20 < 0$; the trace and determinant rule then says nothing about real eigenvalues.

$$\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$$

- The same characteristic polynomial does not make matrices similar. The two matrices below both have characteristic polynomial $(2-\lambda)^2$, but $P^{-1}(2I)P = 2I$ for every P, so only $2I$ is similar to $2I$.

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \qquad \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

- Similarity is not row equivalence: those same two matrices are row equivalent (both reduce to I), but they are not similar.

§3 Homework this week

HW6, Sections 4.4 and 4.6, and HW7, Sections 5.1–5.2, with a one-line hint for each exercise.

Section 4.4.

- **#1, #3** (find $\mathbf{x}$ from $[\mathbf{x}]_{\mathcal{B}}$). $\mathbf{x} = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$.
- **#7** (find $[\mathbf{x}]_{\mathcal{B}}$). Row reduce $[\,\mathbf{b}_1\ \mathbf{b}_2\ \mathbf{b}_3\ \ \mathbf{x}\,]$.
- **#13** (coordinates of a polynomial). Match the coefficients of $1, t, t^2$ and solve the system (P2(a)).
- **#15, #17, #19** (true/false). The definition of $[\mathbf{x}]_{\mathcal{B}}$; $[\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}$ (third trap); $\dim \mathbb{P}_3 = 4$.
- **#25** (the matrix of the coordinate map). It is $P_{\mathcal{B}}^{-1}$.
- **#36** (polynomials as a basis; coordinates of q). Coordinate vectors in the standard basis, then the basis theorem; then solve for the weights.

Section 4.6.

- **#1, #5, #6** (abstract bases). The columns are the coordinates of the old basis vectors in the new basis; then multiply (abstract example).
- **#9** (bases of $\mathbb{R}^2$). Row reduce $[\,\mathbf{c}_1\ \mathbf{c}_2 \mid \mathbf{b}_1\ \mathbf{b}_2\,]$; the other matrix is its inverse.
- **#12, #13** (true/false). The columns are coordinate vectors of an independent set; with $\mathcal{C}$ standard, $P_{\mathcal{C}\leftarrow\mathcal{B}} = P_{\mathcal{B}}$.
- **#15** (polynomial bases). Coefficient vectors as columns (the $\mathbb{P}_2$ example).

Section 5.1.

- **#3, #5** (is it an eigenvector/eigenvalue?). Compute $A\mathbf{u}$ and compare with $\mathbf{u}$.
- **#7** (is 4 an eigenvalue?). Row reduce $A - 4I$ and look for a free variable.
- **#9, #13** (eigenspace bases). Parametric vector form of $(A-\lambda I)\mathbf{x} = \mathbf{0}$.
- **#23, #24, #27, #28, #29, #30** (true/false). 0 eigenvalue $\iff$ singular; $A\mathbf{x} = \lambda\mathbf{x}$ is $(A-\lambda I)\mathbf{x} = \mathbf{0}$; the last trap of 5.1; Theorem 1 needs a triangular matrix; $A(2\mathbf{v}) = 2(A\mathbf{v})$; the eigenspace is a null space.
- **#34** ($A^2 = 0$). If $A\mathbf{x} = \lambda\mathbf{x}$ then $\mathbf{0} = A^2\mathbf{x} = \lambda^2\mathbf{x}$.

Section 5.2.

- **#5, #7** (2×2 characteristic polynomials). $\lambda^2 - (\operatorname{tr} A)\lambda + \det A$; for #7 look at the discriminant.
- **#11, #13** (3×3). Expand along a row or column with zeros and keep the factors.
- **#15** (triangular). Read the diagonal, with multiplicities.
- **#18** (choose h). The eigenspace for 5 is $\operatorname{Nul}(A-5I)$; choose h so that $A-5I$ has two free variables.
- **#19** (det A and the eigenvalues). Set $\lambda = 0$ in the factored characteristic polynomial.
- **#24, #25** (true/false). Theorem 4; the characteristic equation.

§4 Problem set

P1. Find $[\mathbf{x}]_{\mathcal{B}}$ for $\mathbf{x} = (4, 9)$ and $\mathcal{B} = \{(1, 3), (2, 5)\}$.

P2. (a) $\mathcal{B} = \{1 + t,\ t + t^2,\ 1 + t^2\}$ is a basis of $\mathbb{P}_2$. Find $[p]_{\mathcal{B}}$ for $p = 3 + 2t - t^2$. (b) Use coordinate vectors to decide whether $1 - t$, $t - t^2$, $1 - t^2$ are linearly independent.

P3. $\mathcal{B} = \{(1, 2), (3, 4)\}$ and $\mathcal{C} = \{(1, 1), (0, 1)\}$. Find $P_{\mathcal{C} \leftarrow \mathcal{B}}$ and $P_{\mathcal{B} \leftarrow \mathcal{C}}$. If $[\mathbf{x}]_{\mathcal{B}} = (2, -1)$, find $[\mathbf{x}]_{\mathcal{C}}$.

P4. For the matrix below, is $\mathbf{v} = (1, -1, 0)$ an eigenvector? If so, for which eigenvalue? Is 4 an eigenvalue?

$$A = \begin{bmatrix} 3 & 0 & 1 \\ -1 & 2 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

P5. For the same A, find the characteristic polynomial, all eigenvalues, and a basis of the eigenspace for $\lambda = 1$.

P6. Find the characteristic polynomials and eigenvalues of the matrices below.

$$\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix} \qquad \begin{bmatrix} 5 & -3 \\ -4 & 3 \end{bmatrix}$$

P7. List the eigenvalues, with multiplicities, of the matrix below. Is it invertible?

$$\begin{bmatrix} 2 & 3 & -1 & 4 \\ 0 & -1 & 5 & 2 \\ 0 & 0 & 2 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

P8. Suppose $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$. (a) Show that $\mathbf{x}$ is an eigenvector of A^2 and of $A + 3I$; for which eigenvalues? (b) If A is invertible, show $\lambda \neq 0$ and $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$.